week 7

02.03.2026

Cake cutting

① A cake cutting algorithm is strategyproof if on every instance no agent can improve their allocation by answering the algorithm's queries non-truthfully.

Is Cut-and-Choose strategyproof?

chocolate	vanilla

- agent 1 likes the entire cake uniformly
- agent 2 likes vanilla only
- if agent 1 knows that agent 2 likes vanilla only, she can cut the cake at

$$\boxed{3/4 - \varepsilon}$$

Other algorithms we have seen are not strategyproof either.

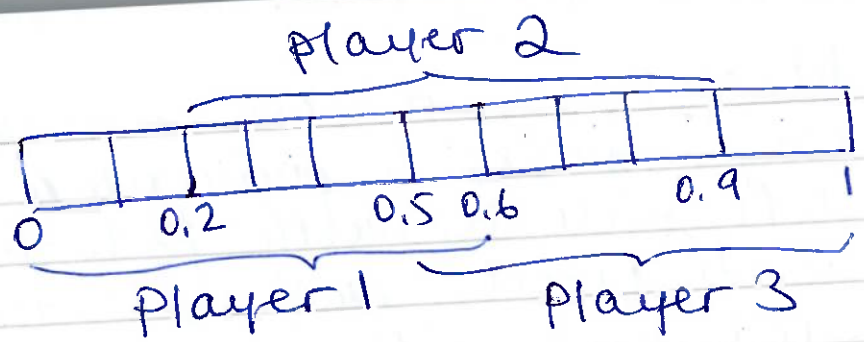
Theorem: there exists no strategyproof algorithm for 2 agents that is Pareto-optimal and envy-free.

Theorem: for any k, one cannot find an allocation of the cake that maximises the sum of agents' utilities in the Robertson-Webb model using k queries, even for $n = 2$.

② Dubins-Spanier (moving knife) procedure

Applicable to any number of agents.

At each stage, ask remaining players to mark their $1/n$ valuation point, and assign the piece to the player that has the leftmost mark.



We use the cake $[0,1]$. Each player values only a sub-interval, uniformly there, total value 1:

- Player 1: values $[0, 0.6]$, density $5/3$, 0 elsewhere
- Player 2: values $[0.2, 0.9]$, density $1/0.7$, 0 elsewhere
- Player 3: values $[0.5, 1]$, density 2, 0 elsewhere

So each $v_i([0,1]) = 1$.

At each round, every remaining player marks a point where the left piece is worth 1/3 to them; the leftmost mark gets the piece and exits.

Round 1 (players 1, 2, 3 on)

Each wants value 1/3

- P1: only $[0, 0.6]$.
  Value of $[0, x]$ is $(5/3)x$.
  $(5/3)x = 1/3 \Rightarrow x = 0.2$.
  $\boxed{\text{Mark at } x_1 = 0.2}$

- P2: only $[0.2, 0.9]$
  From 0.2 to x: value
  $\frac{1}{0.7}(x - 0.2)$.
  $\frac{1}{0.7}(x - 0.2) = 1/3 \Rightarrow$
  $x - 0.2 = \frac{0.7}{3} \Rightarrow x \simeq 0.4(3)$
  $\boxed{\text{Mark } x_2 \simeq 0.4(3)}$

- P3: only $[0.5, 1]$. From 0.5 to x:
value $2(x-0.5)$
$2(x-0.5) = 1/3 \Rightarrow$
$x = 0.5 + 1/6 \approx 0.(6)$
$\boxed{\text{Mark } x_3 \approx 0.(6).}$

Leftmost mark is P1 at 0.2, so P1 gets $[0, 0.2]$ and exits.
Value to P1: $\frac{5}{3} \cdot 0.2 = \frac{1}{3}$.

Remaining cake: $[0.2, 1]$
Remaining players: 2, 3.

<u>Round 2 (players 2,3 on $[0.2, 1]$)</u>
Again, each marks a $1/3$-value point.
Solving as before, $x_2 \approx 0.4(3)$
$x_3 = 0.(6)$

Leftmost mark is P2 at $\approx 0.4(3)$
P2 gets $[0.2, 0.4(3)]$.

Value to P2: intersection with $[0.2, 0.9]$ is itself, length $\simeq 0.2(3)$. With density $\frac{1}{0.7}$,

value $\simeq \frac{0.2(3)}{0.7} = \frac{1}{3}$.

Remaining cake $[0.4(3), 1]$, last player P3 gets it.

Value to P3: intersection with $[0.5, 1]$ is $[0.5, 1]$, length 0.5, value $2 \cdot 0.5 = 1$.

<u>Proportionality and envy</u>:

- Each player gets at least $1/3$: P1 – $1/3$, P2 – $1/3$, P3 – 1, so the allocation is proportional.

- From P1's perspective:
  - their piece is $[0, 0.2]$, value $1/3$.

- P2's piece is $[0.2, 0.4(3)]$, length $0.2(3)$, value to P1
  $5/3 \cdot 0.2(3) = 0.3(8) > \frac{1}{3}$
  P1 envies P2
- P3's piece includes $[0.5, 0.6]$, length 0.1, value $5/3 \cdot 0.1 = \frac{1}{6}$
  So P1 values P3's piece as $1/6 < 1/3$, no envy.

- From P2's perspective
  - no envy to P1 (their piece does not intersect $[0.2, 0.9]$)
  - P3's piece: $[0.4(3), 1]$ intersects $[0.2, 0.9]$ in $[0.4(3), 0.9]$, length $= 0.4(6)$, value $= \frac{0.4(6)}{0.7} = \frac{2}{3}$
    So P2 strictly prefers P3's piece and envies P3.
- P3 doesn't envy anyone as P1's and P2's pieces do not intersect $[0.5, 1]$.

$\Rightarrow$ Dubins-Spanier is proportional but not envy-free.

③ <u>Even-Paz (divide and conquer) procedure</u>

Cut around the median of players' 1/2 marks, giving left half to some players and right half to others, then recurse.

$\boxed{\text{SAME EXAMPLE}}$

Each player marks where they see half the cake:

- P1: needs value 1/2 on $[0, 0.6]$
  $(5/3)\,x = 1/2 \Rightarrow x = 0.3$
- P2: wants $\frac{1}{0.7}(x - 0.2) = 1/2$
  $\Rightarrow x - 0.2 = 0.35 \Rightarrow x = 0.55$
- P3: $2(x - 0.5) = 1/2 \Rightarrow x = 0.75$

Median of $\{0.3, 0.55, 0.75\}$ is 0.55.
We cut at 0.55 and assign players with "small" marks to the left and the rest to the right. Specifically:

- Left group $L = \{P1, P2\}$ on $[0, 0.55]$
- Right group $R = \{P3\}$ on $[0.55, 1]$ (P3 alone gets that piece).

Check P3's value: intersection $[0.55, 1] \cap [0.5, 1] = [0.55, 1]$, length 0.45, value $2 \cdot 0.45 = 0.9 > \frac{1}{3}$.

Now recursively divide $[0, 0.55]$ between P1 and P2 (2-player proportional algorithm, we can just use cut-and-choose):

- P1 cuts [0, 55] into two equal-valued pieces for them. They value only up to $0.55 \wedge 0.6 = 0.55$.

  Value of $[0, x]$ is $(5/3)\,x$; to get half of their value for $[0, 0.55]$ (which is $(5/3) \cdot 0.55$), we cut at midpoint 0.275. So pieces:

  $[0, 0.275]$ and $[0.275, 0.55]$

- P2 chooses between these two. P2's valued interval in $[0, 0.55]$ is $[0.2, 0.55]$

Compute P2's values:

- $[0, 0.275]$ contributes $[0.2, 0.275]$ length 0.075 value $0.075/0.7 \simeq 0.1071$

- $[0.275, 0.55]$ contributes $[0, 275, 0.55]$ length 0.275, value $0.275/0.7 \approx 0.3929$

So P2 picks $[0.275, 0.55]$ and P1 gets $[0, 0.275]$.

Final allocation:

- P1 : $[0, 0.275]$
  Value: $(5/3) \cdot 0.275 \approx 0.4583 > \frac{1}{3}$
- P2: $[0.275, 0.55]$
  valued part $[0.275, 0.55] \cap [0.2, 0.9]$ is length 0.275, value $\approx$ $0.275/0.7 = 0.3929 > \frac{1}{3}$
- P3: $[0.55, 1]$, value 0.9 as computed.

$\Rightarrow$ propoptional.

Check envy:

- P1's view:
  - own $[0, 0.275]$, value ≈ 0.4583

- P2's : [0,275, 0.55] intersect [0, 0.6] => same interval, length 0.275, value also ≈ 0.4583 (P1 sees them equal)
- P3's: [0.55, 1] intersects [0, 0.6] at [0.55, 0.6], length 0.05, value $(5/3) \cdot 0.05 = 0.0833$

=> P1 does not envy anyone (indifferent with P2 and prefers their own to P3's)

- P2' view:
  - own value ≈ 0.3929
  - P1's: [0, 0.275] intersects [0.2, 0.9] at [0.2, 0.275], length 0.075, value ≈ 0.1021
  - P3's [0.55, 1] intersect [0.2, 0.9] is [0.55, 0.9], length 0.35, value ≈ 0.35/0.7 = 0.5

  so P2 envies P3 (0.5 > 0.3929)

$\Rightarrow$ Even-Paz yields a proportional but not envy-free allocation.

(4) Selfridge-Conway
(3 players, envy-free)

We apply the Selfridge-Conway procedure to the same instance.

Step 1: P1 cuts the cake into 3 equal pieces (for them)

We already did this above: P1's thirds at 1/3 increments of their value over [0, 0.6].

P1 cuts at 0.2 and 0.4:

- $X = [0, 0.2]$
- $Y = [0.2, 0.4]$
- $Z = [0.4, 1]$

Each worth exactly 1/3 to P1.

<u>Step 2</u>: P2 trims the largest piece.

Compute P2's values:

- $v_2(X) = 0$ (no intersection with $[0.2, 0.9]$)
- $v_2(Y) = \frac{1}{0.7} \cdot 0.2 \simeq 0.2857$
- $v_3(Z) = \frac{1}{0.7} \cdot 0.5 \simeq 0.7143$

Largest is Z. P2 trims Z from the left to make its value equal to $v_2(Y)$.

Set $Z' = [t, 1]$ such that
$$v_2(Z') = v_2(Y).$$

Valued part of Z' is $[t, 0.9]$, length $0.9 - t$.

Solve:

$$\frac{1}{0.7}(0.9 - t) = \frac{1}{0.7} \cdot 0.2 \Rightarrow 0.9 - t = 0.2$$
$$\Rightarrow t = 0.7$$

So main trimmed piece $Z' = [0.7, 1]$, trimmings $T = [0.4, 0.7]$.

We now have three main pieces: X, Y, Z', plus trimmings T.

<u>Step 3:</u> P3 chooses first among X, Y, Z'; P2 second; P1 last

Compute P3's values:

- $v_3(X) = 0$ (all left of 0.5)
- $v_3(Y) = 0$
- $v_3(Z') = [0.7, 1] \cap [0.5, 1] =$
  $= [0.7, 1]$, length 0.3,
  value $2 \cdot 0.3 = 0.6$

So P3 takes Z'.

Because P3 took the trimmed piece, P2 is now free to choose between X and Y (no forced choice). P2 values:

- $v_2(X) = 0$
- $v_2(Y) \simeq 0.2857$

So P2 takes $Y = [0.2, 0.4]$;
P1 gets $X = [0, 0.2]$.

<u>Step 4:</u> Dividing the trimmings.

By Selfridge-Conway:

- Between P2 and P3, P3 took the trimmed piece, P2 an untrimmed piece.
- So P2 cuts T into 3 pieces of equal value to them, then trimmings are chosen in order: first P3, then P1, then P2.

~~P3 values only $[0.5, 1]$; within $T = [0.4, 0.7]$ they value $[0.5, 0.7]$, length 0.2, total value 0.4. Each trimming piece would be worth 0.4/3 to P3, so each has valued length $0.2/3 \approx 0.06$~~

P2 values $T = [0.4, 0.7]$ uniformly, so it divides T into three parts

$T_1 = [0.4, 0.5]$
$T_2 = [0.5, 0.6]$
$T_3 = [0.6, 0.7]$

each of the same value $0.1/0.7 = 1/7$ to them.

Then:

- P_3 picks their favourite among T_1, T_2, T_3: they value T_1 at 0 and are indifferent between T_2 and T_3 – assume they take T_3.
- P1 picks their favourite among the remaining two; they value T_1, T_2 equally and T_3 at 0, assume they take T_1;

So P2 takes $Y = [0.2, 0.4]$;
P1 gets $X = [0, 0.2]$.

<u>Step 4:</u> Dividing the trimmings.

By Selfridge-Conway:

- Between P2 and P3, P3 took the trimmed piece, P2 an untrimmed piece.
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~~P3 values only $[0.5, 1]$; within $T = [0.4, 0.7]$ they value $[0.5, 0.7]$, length 0.2, total value 0.4. Each trimming piece would be worth 0.4/3 to P3, so each has valued length $0.2/3 \approx 0.066$~~

- P2 gets the last (T_2).

Check envy-freeness

- P3 got to choose their favourite piece among both the main pieces and the trimmings, so they do not envy anyone.

- P2 cut the trimmings into three equal pieces to them and got one of the largest main pieces (Y), so there is no envy either.

- P1 ranked the three original pieces as equal. After trimming, they get a main piece that in their view is at least as good as what remains plus a trimming, so they do not envy either.